Affective Temperaments in High-Security Forensic Psychiatric Inpatients: A Cross-Sectional Comparison With Psychiatric Outpatients and Healthy Controls

Abstract

Objective: To compare affective temperament in high-security forensic psychiatric inpatients, psychiatric outpatients and healthy controls.

Methods: This cross-sectional study included 66 forensic inpatients, 25 outpatients and 24 healthy controls, all men. Participants completed the Turkish TEMPS-A. Group comparisons used one-way ANOVA with Tukey post hoc tests and chi-square tests.

Results: Depressive, cyclothymic, irritable and anxious scores differed significantly across groups. The hyperthymic comparison was not significant. Forensic inpatients had higher irritable scores than outpatients (11.58 ± 4.78 versus 8.12 ± 4.00; Tukey p = .002) and lower depressive scores (10.09 ± 4.27 versus 12.68 ± 4.29; p = .020). The two patient groups did not differ significantly in the other three dimensions. Both patient groups scored higher than controls in depressive, cyclothymic, irritable and anxious temperament.

Conclusion: Forensic inpatients had higher irritable and lower depressive temperament scores than psychiatric outpatients. These exploratory, unadjusted comparisons describe the sampled groups and do not establish temperament as a cause or predictor of offending.

Keywords: Affective temperament; Forensic psychiatry; TEMPS-A; Schizophrenia; Cross-sectional study

Introduction

Psychiatric diagnosis alone does not explain differences in violence-related outcomes. A systematic review found an association between schizophrenia-spectrum disorders and violence perpetration (Whiting et al., 2022), while a review of longitudinal studies identified previous offending and substance misuse as important risk markers (Lagerberg et al., 2025). These associations do not imply that every person with a psychotic disorder is violent. In a review restricted to physical violence in non-affective psychosis, associations with education and psychopathy did not remain significant after correction for multiple comparisons (Ahle et al., 2026). These differences show why the definition of violence and the analytical approach matter.

Offending, arrest, aggressive behavior and recidivism are distinct outcomes. Historical research in sheltered care found different arrest patterns across crime categories (Hwang & Segal, 1996), and clinical reviews describe varied correlates of aggression and offending (Ayar & Ayar, 2021; Pompili et al., 2017). Turkish studies have described diagnostic and offense characteristics in forensic referrals and selected samples involving serious offenses (Türkcan et al., 2000; İnan et al., 2018). Findings from these selected populations cannot establish crime rates in the wider population or an individual's propensity for violence.

Affective temperament describes characteristic patterns of emotional experience and response. Its study developed partly through research on mood disorders and semi-structured temperament interviews (Cassano et al., 1989; Placidi et al., 1998). TEMPS-A assesses depressive, cyclothymic, hyperthymic, irritable and anxious dimensions. The Turkish version was evaluated in a psychometric study by Vahip et al. (2005). These dimensions are distinct from current symptoms and character traits. Research on psychopathic subtypes and legal cynicism examines related but different constructs and is not direct evidence about TEMPS-A scores (Martínez-López et al., 2019; Ameri et al., 2019).

Temperament differences have been examined in schizophrenia and in relation to aggression in bipolar disorder (Dib et al., 2021; Fico et al., 2023). However, temperament ratings, aggression measures and forensic involvement represent different constructs. In a Turkish study of patients attending forensic and general psychiatric outpatient clinics, temperament scores did not differ significantly between patients with and without offense-related judicial involvement (Çatlı et al., 2026). Including both psychiatric outpatients and healthy controls may help distinguish differences associated with psychiatric illness from those observed between forensic inpatients and outpatients.

This study compared high-security forensic psychiatric inpatients with psychiatric outpatients and healthy controls across the five affective temperament dimensions. Comparisons with healthy controls provided context for the differences between the two patient groups. The aim was descriptive, not to develop a violence prediction tool or establish a causal relationship with offending. Irritability was not a prespecified primary outcome.

Methods

Design, setting and ethics

This cross-sectional study was conducted between December 1, 2024, and February 28, 2025, at the High-Security Forensic Psychiatry (YGAP) Unit and Psychiatry Outpatient Clinic of Adana City Training and Research Hospital. The study followed the Declaration of Helsinki and received approval from the hospital ethics committee on November 7, 2024 (approval no. 236). All participants provided their own written informed consent.

Participants and recruitment

The study included 66 forensic psychiatric inpatients, 25 psychiatric outpatients and 24 healthy controls. All participants were men. The forensic group was recruited first. The other groups were selected to be similar in age, education and sex. Patients were recruited from the forensic unit and ongoing psychiatric outpatient care. The diagnostic categories were paranoid schizophrenia, other schizophrenia subtypes and atypical psychosis, and bipolar affective disorder. Diagnoses were based on clinical records and assessment by the supervising psychiatrist.

Patients were clinically stable, as judged by a psychiatrist, and able to participate in the assessment. Agitated patients were not included. Participants had to be literate and free of severe hearing or visual impairment. Intellectual disability or cognitive impairment that prevented comprehension and completion of the questionnaire was an exclusion criterion, as was the absence of written consent. Standardized clinical symptom scales were not administered.

Approximately 90 patients were in the forensic unit during the study period. Some could not complete the questionnaire because of intellectual disability or severe cognitive impairment associated with psychosis.

Forensic inpatients had court-related admissions following offenses such as homicide, assault, theft and property damage. Offense histories were assessed using records and participant questioning. Both violent and nonviolent offenses were represented, so forensic group membership was not a direct measure of violence severity. Controls were recruited from hospital employees, including cleaning and support staff, and screened through record review and self-report. All individuals approached agreed to participate.

Measures

A semi-structured form recorded age, education, marital status, employment, social security coverage and diagnosis in patients. For forensic inpatients, employment status referred to the period before admission; employment was assessed with reference to clinically stable periods. Although groups were selected to be similar in education, their educational distributions were not identical.

Affective temperament was assessed using the 99-item Turkish TEMPS-A. The subscales comprise 18 depressive, 19 cyclothymic, 20 hyperthymic, 18 irritable and 24 anxious items. Participants answer yes or no based on their usual characteristics over much of their lives. The Turkish psychometric study and subsequent corrigendum were published by Vahip et al. (2005, 2006). Subscale scores were analyzed as continuous variables, without classifying participants by dominant temperament.

One researcher administered the questionnaire in the forensic unit. Some participants completed it independently, while others received reading assistance. No missing or multiple item responses were reported. Internal consistency was not estimated in this sample.

Statistical analysis

The original analyses were conducted using IBM SPSS Statistics version 22.0. During revision, descriptive statistics, one-way ANOVA, Tukey comparisons and Pearson chi-square tests were recalculated from the supplied dataset using Python 3.12.14 and SciPy 1.17.0. Continuous variables are presented as means and standard deviations, and categorical variables as counts and percentages. The original analysis assessed distributions using the Kolmogorov–Smirnov test. Continuous variables were compared across groups using one-way analysis of variance, followed by Tukey post hoc tests. Statistical significance was set at $p < .05$.

Tukey adjustment was applied to pairwise comparisons within each temperament dimension, but no additional adjustment was made across the five dimensions. Findings across dimensions are therefore interpreted as exploratory. Categorical variables were compared using Pearson’s chi-square test. Sparse expected cells in the education and marital-status tables require caution when interpreting these approximate p-values.

Results

Participant characteristics

The sample comprised 66 forensic psychiatric inpatients, 25 psychiatric outpatients and 24 healthy controls. Mean ages were 39.32 ± 10.68, 39.72 ± 8.32 and 37.00 ± 7.27 years, respectively ($F(2, 112) = 0.626$, $p = .537$).

Participant characteristics are presented in Table 1. Education and marital-status distributions differed across groups ($\chi^2(6) = 13.720$, $p = .033$; $\chi^2(4) = 30.889$, $p < .001$). Employment also differed ($\chi^2(2) = 15.238$, $p < .001$). During the assessed period, 41 forensic inpatients (62.1%), 13 outpatients (52.0%) and all 24 controls were employed. The remaining 25 forensic inpatients and 12 outpatients were not employed. Social security coverage was reported for 35 forensic inpatients (53.0%), 15 outpatients (60.0%) and all controls ($\chi^2(2) = 17.188$, $p < .001$). These overall tests do not identify specific pairwise differences in demographic characteristics.

In forensic inpatients, 28 (42.4%) had paranoid schizophrenia, 29 (43.9%) had other schizophrenia subtypes or atypical psychosis, and 9 (13.6%) had bipolar affective disorder. The corresponding outpatient counts were 9 (36.0%), 13 (52.0%) and 3 (12.0%) (Table 2).

Affective temperament

Overall group differences were found in depressive ($F(2, 112) = 32.126$), cyclothymic ($F(2, 112) = 16.418$), irritable ($F(2, 112) = 44.248$) and anxious temperament ($F(2, 112) = 24.966$), all $p < .001$. The hyperthymic comparison was not statistically significant ($F(2, 112) = 2.568$, $p = .081$). Subscale means and standard deviations are presented in Table 3.

Forensic inpatients had higher irritable scores than outpatients (11.58 ± 4.78 versus 8.12 ± 4.00; Tukey $p = .002$) and lower depressive scores (10.09 ± 4.27 versus 12.68 ± 4.29; $p = .020$). The standard errors of the pairwise differences were 0.978 and 0.952, respectively. The two patient groups did not differ significantly in cyclothymic, hyperthymic or anxious temperament.

Both patient groups scored higher than controls in depressive, cyclothymic, irritable and anxious temperament (each Tukey $p < .001$). None of the pairwise comparisons of hyperthymic scores was significant. A nonsignificant result does not establish equivalence.

Discussion

The main finding was that forensic inpatients had higher irritable and lower depressive temperament scores than psychiatric outpatients. Both patient groups also scored higher than healthy controls in four temperament dimensions. These comparisons were unadjusted. The differences between the patient groups remain exploratory because no primary dimension was prespecified and the analysis did not correct for comparisons across all five dimensions.

The comparisons with healthy controls partly agree with those of Dib et al. (2021), who studied long-term schizophrenia inpatients and healthy controls. Their patients had higher scores in four dimensions, while controls had higher hyperthymic scores. Our nonsignificant hyperthymic result does not fully replicate that pattern. In bipolar disorder, Fico et al. (2023) reported higher depressive, irritable and anxious scores in aggression-related clusters. Those clusters were derived from both temperament and aggression measures, so the finding does not independently demonstrate an association between temperament and offending.

A Turkish study offers a contrasting finding. Çatlı et al. (2026) found no significant differences in the five temperament dimensions between patients with schizophrenia with and without offense-related judicial involvement. Both groups were recruited in outpatient settings in 2016, unlike our forensic inpatient group. Differences in setting, recruitment, diagnostic composition and clinical characteristics may help explain the contrasting results, but the available evidence cannot determine which factors account for them.

Our comparison of forensic inpatients with psychiatric outpatients combines differences in legal status and treatment setting. A retrospective study comparing forensic and general psychiatric inpatients with schizophrenia-spectrum disorders found differences in clinical and other aggression-related characteristics (Kirchebner et al., 2023). Without a general psychiatric inpatient comparison group, we cannot separate hospitalization from forensic involvement. Criminal history and substance misuse are additional factors to consider (Lagerberg et al., 2025). Fischer-Vieler et al. (2023) also found associations between a history of severe violence and some, but not all, components of clinical insight. These factors were not controlled for in our temperament comparisons.

Irritable temperament may be relevant to emotion regulation, but this study did not assess a mechanism linking them. In qualitative interviews, forensic patients with psychosis described

perceived threat and difficulty managing intense emotions when discussing interpersonal violence (Lambe et al., 2024). Studies of longitudinal impulsivity profiles (Wang et al., 2024) and experimental inhibitory control (Bielecki et al., 2024) used different measures. These findings suggest questions for future research, but irritable TEMPS-A scores should not be treated as measures of impulsivity, loss of control or violent behavior.

The higher depressive temperament score in outpatients does not indicate a current depressive episode. Research on depressive symptoms during antipsychotic treatment examines a different construct (Reynolds & McGowan, 2021), and medication effects cannot be inferred from our findings. Affective temperament has also been studied in euthymic bipolar disorder (Kesebir et al., 2005). However, the mixed diagnoses and small patient groups in this study limit diagnosis-specific interpretation. The nonsignificant hyperthymic comparison does not rule out a difference within the bipolar subgroup.

Clinical and legal decisions require more evidence than an unadjusted cross-sectional association. Research on reconviction after discharge (Bailey & MacCulloch, 1992), community outcomes after homicide (Ong et al., 2009), and treatment or punishment of drug-using offenders (Tomaz et al., 2023) addresses longitudinal outcomes that this study did not assess. A systematic review of violence-risk instruments found mixed evidence of predictive performance (Ogonah et al., 2023). Our findings do not justify using TEMPS-A to predict individual dangerousness, determine discharge or support legal restrictions. Empirical findings must also be distinguished from legal and ethical judgments in forensic assessment (Gkotsi & Gasser, 2016; Aydoğdu & Saka, 2024).

Limitations

The cross-sectional design prevents conclusions about temporal order, causation or prediction. The single-center, all-male sample limits generalizability. Excluding agitated patients and those unable to complete the questionnaire means that the findings do not represent the full forensic inpatient population. Group sizes were unequal. Controls were hospital employees and may differ from patients in employment and other social characteristics. Selecting groups for similar age and education does not eliminate confounding.

Clinical stability was judged by a psychiatrist without standardized symptom ratings. Retrospective self-report and the use of both independent and reading-assisted questionnaire

completion may have affected responses. Internal consistency was not estimated in this sample. Testing multiple temperament dimensions increases the chance of false-positive findings beyond the comparisons covered by the within-dimension Tukey adjustment. Sparse expected cells also limit the categorical chi-square analyses. Future violence, offense severity and recidivism were not assessed as outcomes.

Effect sizes and confidence intervals are not reported, which limits assessment of the magnitude and precision of the differences. Future studies should prespecify the main comparison, measure relevant clinical covariates and report effect sizes with confidence intervals to aid interpretation beyond statistical significance (Lakens, 2013).

Conclusion

High-security forensic psychiatric inpatients had higher irritable and lower depressive temperament scores than psychiatric outpatients. These findings describe the sampled groups and need further study with better control of clinical differences and treatment setting. They do not establish temperament as a cause, independent risk factor or predictor of offending.

Declarations

Data availability: Participant-level data are not publicly available because of confidentiality restrictions. Requests for access to study data or supporting records for journal review may be directed to the corresponding author and will be considered subject to applicable confidentiality and access requirements.

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References

Ahle, G., Henriksen, M. G., Mølstrøm, I. M., Rasmussen, A. R., Christensen, M. B., & Nordgaard, J. (2026). Predictors of violence against others in non-affective psychosis: A systematic review and meta-analysis. Schizophrenia Research, 291, 68–79. https://doi.org/10.1016/j.schres.2026.02.015

Ameri, T., Burgason, K. A., DeLisi, M., Heirigs, M. H., Hochstetler, A., & Vaughn, M. G. (2019). Legal cynicism: Independent construct or downstream manifestation of antisocial constructs? New evidence. International Journal of Law and Psychiatry, 64, 211–218. https://doi.org/10.1016/j.ijlp.2019.04.008

Ayar, E., & Ayar, Y. (2021). Ruhsal bozukluklarda suça yönelim: Derleme. Abant Sağlık Bilimleri ve Teknolojileri Dergisi, 1(2), 83–91. https://dergipark.org.tr/tr/pub/sabited/article/1074174

Aydoğdu, H. İ., & Saka, N. E. (2024). Adli psikiyatrik değerlendirmelerde; tehlikelilik kavramına, güvenlik tedbiri kararlarında yasal kısıtlılık ve zorunlu yatışlara evrensel bakış. In G. Oral (Ed.), Dinçmen Adli Psikiyatri (pp. 130–138). İstanbul Üniversitesi–Cerrahpaşa Üniversite Yayınevi. https://doi.org/10.5152/8214

Bailey, J., & MacCulloch, M. (1992). Patterns of reconviction in patients discharged directly to the community from a special hospital: Implications for aftercare. The Journal of Forensic Psychiatry, 3(3), 445–461. https://doi.org/10.1080/09585189208409021

Bielecki, M., Tyburski, E., Plichta, P., Samochowiec, J., Kucharska-Mazur, J., Podwalski, P., Rek-Owodziń, K., Waszczuk, K., Sagan, L., Michalczyk, A., Rudkowski, K., Karabanowicz, E., Świątkowska, K., Misiak, B., Bąba-Kubiś, A., & Mak, M. (2024). Impulsivity and inhibitory control in deficit and non-deficit schizophrenia. BMC Psychiatry, 24, 473. https://doi.org/10.1186/s12888-024-05918-6

Cassano, G. B., Akiskal, H. S., Musetti, L., Perugi, G., Soriani, A., & Mignani, V. (1989). Psychopathology, temperament, and past course in primary major depressions. 2. Toward a redefinition of bipolarity with a new semistructured interview for depression. Psychopathology, 22(5), 278–288. https://doi.org/10.1159/000284608

Çatlı, B., Çakmak, I. B., Gürkan, Ş., Tatlıdil Yaylacı, E., Kaya, H., & Göka, E. (2026). Temperament and insight in offender and non-offender patients with schizophrenia in Türkiye. Psychiatry and Clinical Psychopharmacology, 36(3), 301–311. https://doi.org/10.5152/pcp.2026.251292

Dib, J. E., Nehme, I., Haddad, C., Azar, J., Hallit, S., & Obeid, S. (2021). Affective temperaments of Lebanese patients with schizophrenia: Comparison by gender and severity of psychosis. BMC Research Notes, 14, 430. https://doi.org/10.1186/s13104-021-05854-8

Fico, G., Janiri, D., Pinna, M., Sagué-Vilavella, M., Gimenez Palomo, A., Oliva, V., De Prisco, M., Cortez, P. G., Anmella, G., Gonda, X., Sani, G., Tondo, L., Vieta, E., & Murru, A. (2023). Affective temperaments mediate aggressive dimensions in bipolar disorders: A cluster analysis from a large, cross-sectional, international study. Journal of Affective Disorders, 323, 327–335. https://doi.org/10.1016/j.jad.2022.11.084

Fischer-Vieler, T., Ringen, P. A., Kvig, E., Bell, C., Hjell, G., Tesli, N., Rokicki, J., Melle, I., Andreassen, O. A., Friestad, C., & Haukvik, U. K. (2023). Associations between clinical insight and history of severe violence in patients with psychosis. Schizophrenia Bulletin Open, 4(1), sgad011. https://doi.org/10.1093/schizbullopen/sgad011

Gkotsi, G. M., & Gasser, J. (2016). Neuroscience in forensic psychiatry: From responsibility to dangerousness. Ethical and legal implications of using neuroscience for dangerousness assessments. International Journal of Law and Psychiatry, 46, 58–67. https://doi.org/10.1016/j.ijlp.2016.02.030

Hwang, S. D., & Segal, S. P. (1996). Criminality of the mentally ill in sheltered care: Are they more dangerous? International Journal of Law and Psychiatry, 19(1), 93–105. https://doi.org/10.1016/0160-2527(95)00029-1

İnan, S., Yıldızhan, E., & Öncü, F. (2018). The disease history, sociodemographics, and criminal features of the homicidal forensic psychiatric patients. Türk Psikiyatri Dergisi, 29(4), 258–268. https://pubmed.ncbi.nlm.nih.gov/30887476/

Kesebir, S., Vahip, S., Akdeniz, F., & Yüncü, Z. (2005). Bipolar bozuklukta mizaç ile klinik özelliklerin ilişkisi. Türk Psikiyatri Dergisi, 16(3), 164–169. https://pubmed.ncbi.nlm.nih.gov/16180149/

Kirchebner, J., Lau, S., & Machetanz, L. (2023). Offenders and non-offenders with schizophrenia spectrum disorders: Do they really differ in known risk factors for aggression? Frontiers in Psychiatry, 14, 1145644. https://doi.org/10.3389/fpsyt.2023.1145644

Lagerberg, T., Lambe, S., Paulino, A., Yu, R., & Fazel, S. (2025). Systematic review of risk factors for violence in psychosis: 10-year update. The British Journal of Psychiatry, 226(2), 100–107. https://doi.org/10.1192/bjp.2024.120

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. Frontiers in Psychology, 4, 863. https://doi.org/10.3389/fpsyg.2013.00863

Lambe, S., Cooper, K., Fazel, S., & Freeman, D. (2024). Psychological framework to understand interpersonal violence by forensic patients with psychosis. The British Journal of Psychiatry, 224(2), 47–54. https://doi.org/10.1192/bjp.2023.132

Martínez-López, J. N. I., Medina-Mora, M. E., Robles-García, R., Madrigal, E., Juárez, F., Tovilla-Zarate, C. A., Reyes, C., Monroy, N., & Fresán, A. (2019). Psychopathic disorder subtypes based on temperament and character differences. International Journal of Environmental Research and Public Health, 16(23), 4761. https://doi.org/10.3390/ijerph16234761

Ogonah, M. G. T., Seyedsalehi, A., Whiting, D., & Fazel, S. (2023). Violence risk assessment instruments in forensic psychiatric populations: A systematic review and meta-analysis. The Lancet Psychiatry, 10(10), 780–789. https://doi.org/10.1016/S2215-0366(23)00256-0

Ong, K., Carroll, A., Reid, S., & Deacon, A. (2009). Community outcomes of mentally disordered homicide offenders in Victoria. Australian & New Zealand Journal of Psychiatry, 43(8), 775–780. https://doi.org/10.1080/00048670903001976

Placidi, G. F., Signoretta, S., Liguori, A., Gervasi, R., Maremmani, I., & Akiskal, H. S. (1998). The semi-structured affective temperament interview (TEMPS-I). Reliability and psychometric properties in 1010 14–26-year old students. Journal of Affective Disorders, 47(1–3), 1–10. https://doi.org/10.1016/S0165-0327(97)00122-5

Pompili, E., Carlone, C., Silvestrini, C., & Nicolò, G. (2017). Focus on aggressive behaviour in mental illness. Rivista di Psichiatria, 52(5), 175–179. https://doi.org/10.1708/2801.28344

Reynolds, G. P., & McGowan, O. O. (2021). Schizophrenia, depressive symptoms, and antipsychotic drug treatment. International Journal of Neuropsychopharmacology, 24(4), 253–255. https://doi.org/10.1093/ijnp/pyaa091

Tomaz, V., Moreira, D., & Souza Cruz, O. (2023). Criminal reactions to drug-using offenders: A systematic review of the effect of treatment and/or punishment on reduction of drug use and/or criminal recidivism. Frontiers in Psychiatry, 14, 935755. https://doi.org/10.3389/fpsyt.2023.935755

Türkcan, S., İncesu, C., Canbek, Ö., Can, Y., Sercan, M., & Uygur, N. (2000). 1831 adli olgunun tanı dağılımı ve tanı-suç bağlantısının değerlendirilmesi. Düşünen Adam, 13(3), 132–137. https://dusunenadamdergisi.org/article/886

Vahip, S., Kesebir, S., Alkan, M., Yazici, O., Akiskal, K. K., & Akiskal, H. S. (2005). Affective temperaments in clinically-well subjects in Turkey: Initial psychometric data on the TEMPS-A. Journal of Affective Disorders, 85(1–2), 113–125. https://doi.org/10.1016/j.jad.2003.10.011

Vahip, S., Kesebir, S., Alkan, M., Yazıcı, O., Akiskal, K. K., & Akiskal, H. S. (2006). Corrigendum to “Affective temperaments in clinically-well subjects in Turkey: Initial psychometric data on the TEMPS-A” [J. Affect. Disord. 85 (2005) 113–125]. Journal of Affective Disorders, 92(2–3), 313. https://doi.org/10.1016/j.jad.2006.01.019

Wang, Q. K., Yang, Q., Li, C. X., Qiu, Y. F., Yin, X. T., Hu, J. M., Zhang, Q. T., & Chen, X. C. (2024). Distinct impulsivity profiles in subtypes of violence among community-dwelling patients with severe mental disorders: A longitudinal study. BMC Psychiatry, 24, 590. https://doi.org/10.1186/s12888-024-06044-z

Whiting, D., Gulati, G., Geddes, J. R., & Fazel, S. (2022). Association of schizophrenia spectrum disorders and violence perpetration in adults and adolescents from 15 countries: A systematic review and meta-analysis. JAMA Psychiatry, 79(2), 120–132. https://doi.org/10.1001/jamapsychiatry.2021.3721